CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

Project Report

Submitted by

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1. Introduction

Customer segmentation helps businesses group customers based on behavior and demographics. This project uses K-Means clustering in R to identify distinct customer groups.

2. Problem Statement

To segment customers into groups based on Age, Income, Spending Score, and Purchase Frequency using K-Means clustering and visualize the results.

3.Dataset Description

The dataset is synthetically generated using normal distribution with 200 records and four features.

4.Methodology

Data Generation → Scaling → K-Means Clustering → Visualization

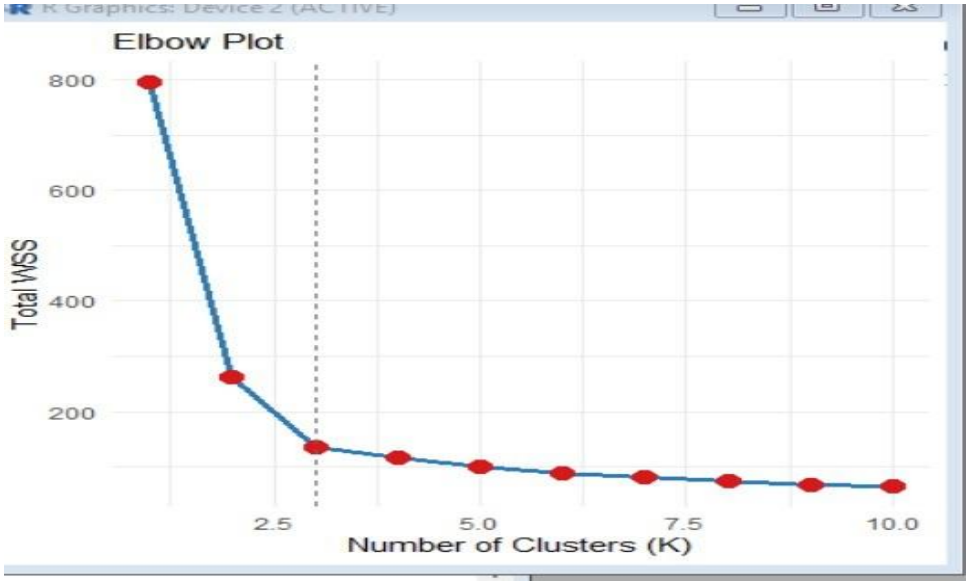
5.Dataset Generation Code

1. Elbow Plot Code and Output

```
library(ggplot2); library(dplyr); library(tidyr)
set.seed(42) n <- 200
customer_data <- data.frame(   Age = c(rnorm(60,25,4),
rnorm(70,45,6), rnorm(70,60,5)),
  Annual_Income = c(rnorm(60,30000,5000), rnorm(70,65000,8000),
rnorm(70,90000,10000)),
  Spending_Score = c(rnorm(60,70,10), rnorm(70,50,12), rnorm(70,30,10)),
```

```
  Purchase_Freq = c(rnorm(60,15,3), rnorm(70,8,2),
rnorm(70,4,1)) )
features_scaled <- scale(customer_data)
```

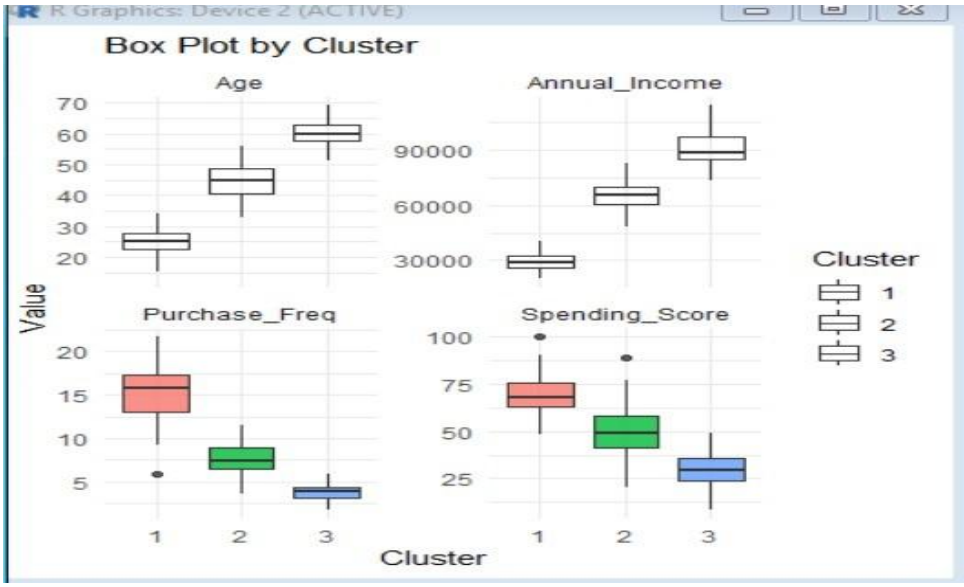
```
wss <- sapply(1:10, function(k) kmeans(features_scaled, k,
nstart=25)$tot.withinss) ggplot(data.frame(K=1:10, WSS=wss), aes(K, WSS)) +
geom_line() +   geom_point() +   geom_vline(xintercept=3, linetype="dashed") +
labs(title="Elbow Plot")
```



2. Box Plot Code and Output

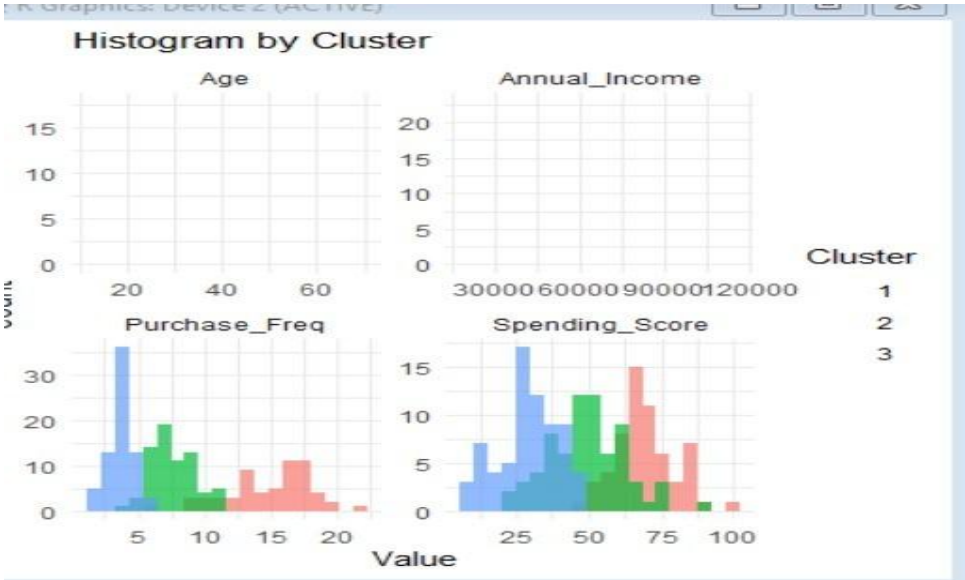
```
customer_data$Cluster <- as.factor(kmeans(features_scaled, 3,
nstart=25)$cluster) customer_data %>%   pivot_longer(-Cluster,
names_to="Feature", values_to="Value") %>%   ggplot(aes(Cluster, Value,
```

```
fill=Cluster)) +   geom_boxplot() +   facet_wrap(~Feature, scales="free_y") +
labs(title="Box Plot by Cluster")
```



3. Histogram Code and Output

```
customer_data %>%   pivot_longer(-Cluster, names_to="Feature",
values_to="Value") %>%   ggplot(aes(Value, fill=Cluster)) +
geom_histogram(bins=20, alpha=0.7) +   facet_wrap(~Feature,
scales="free") +   labs(title="Histogram by Cluster")
```



9. Results and Discussion

Cluster 1 represents young high spenders, Cluster 2 moderate customers, and Cluster 3 older low spenders.

10. Conclusion

K-Means clustering successfully segments customers into meaningful groups for business insights.